Individual proof of completion - Lab 3 - Andrew Ebinger

Task 2:

Question 2 - This action created a subfolder called “webcam”

Step 8

```
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Crystal/lab3/Ebin
ger/roscourse_ws$ colcon build --symlink-install
Starting >>> webcam
--- stderr: webcam
/usr/lib/python3/dist-packages/setuptools/command/easy_install.py:158: EasyInsta
llDeprecationWarning: easy_install command is deprecated. Use build and pip and
other standards-based tools.
  warnings.warn(
---
Finished <<< webcam [1.29s]

Summary: 1 package finished [1.84s]
  1 package had stderr output: webcam
```

Question 9 - The colcon build created the following folders: build, install, and log.

Question 12 - I have the webcam_pub and webcam_sub available to run.

Task 3:

Question 17 - Based on the names, I anticipate that the services will allow me to set the position and color of the turtle using integer values.

Step 22

```
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Crystal/lab3/Ebinger/roscourse_ws/src/turtle_interfa...
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Crystal/lab3/Ebinger/roscourse_ws/src/turtle_interfaces$ ros2 interfa
ce show turtle_interfaces/msg/TurtleMsg
string name
geometry_msgs/Pose turtle_pose
        Point position
                float64 x
                float64 y
                float64 z
        Quaternion orientation
                float64 x 0
                float64 y 0
                float64 z 0
                float64 w 1
string color
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Crystal/lab3/Ebinger/roscourse_ws/src/turtle_interfaces$ ros2 interfa
ce show turtle_interfaces/srv/SetPose
geometry_msgs/PoseStamped turtle_pose
        std_msgs/Header header
                builtin_interfaces/Time stamp
                        int32 sec
                        uint32 nanosec
                string frame_id
        Pose pose
                Point position
                        float64 x
                        float64 y
                        float64 z
                Quaternion orientation
                        float64 x 0
                        float64 y 0
                        float64 z 0
                        float64 w 1
---
int8 ret
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Crystal/lab3/Ebinger/roscourse_ws/src/turtle_interfaces$ ros2 interfa
ce show turtle_interfaces/srv/SetColor
string color
---
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Crystal/lab3/Ebinger/roscourse_ws/src/turtle_interfaces$
```

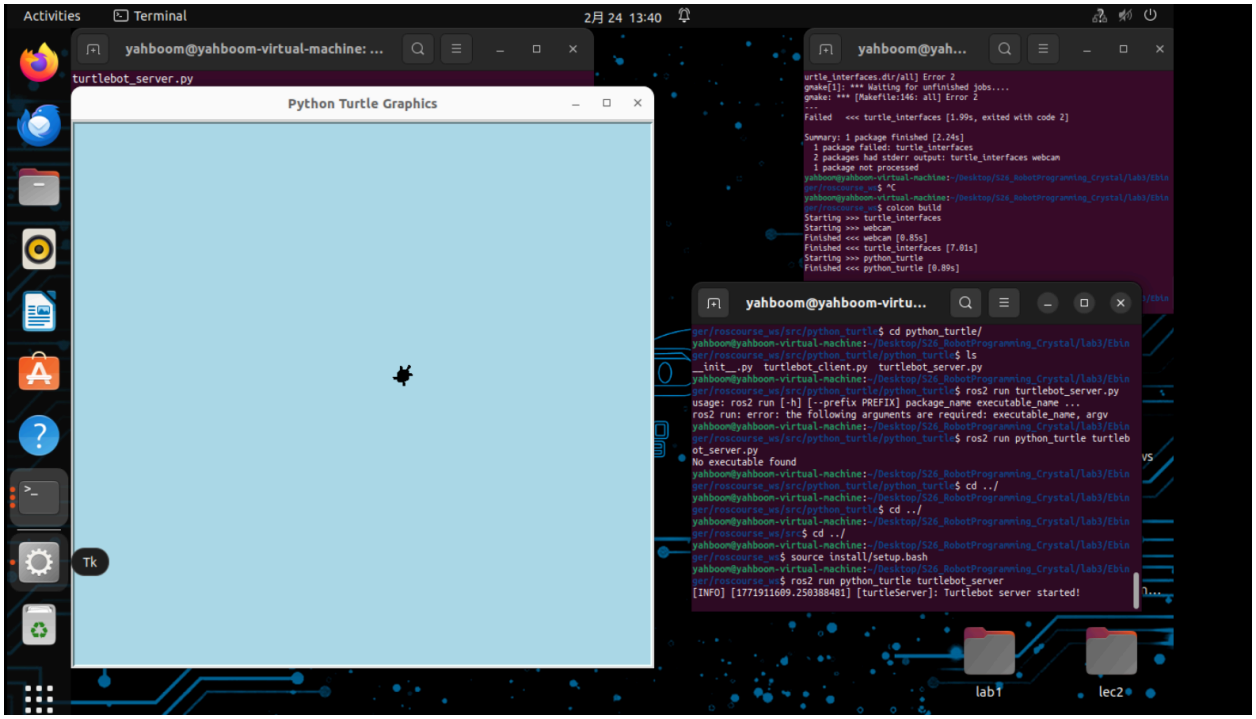
Task 4:

Question 3:

<u>Turtlebot_server:</u> The purpose of Turtlebot server is to compute the properties of Turtlebot during its simulation. The bulk of the file is contained within a class called `TurtlebotServer`, which further contains the functions `twist_callback` and `driving_timer_cb`.
It also has class definitions that create publisher and subscriber nodes for the turtlebot that transmit information about its state to the simulation. `Twist_callback` is fairly simple, only sending calculated linear and angular velocities of the turtlebot to the simulation. `Driving_timer_cb` is more complex, containing not only code to simulate the movement physics of turtlebot but also lines that convert between quaternion and RPY notation.

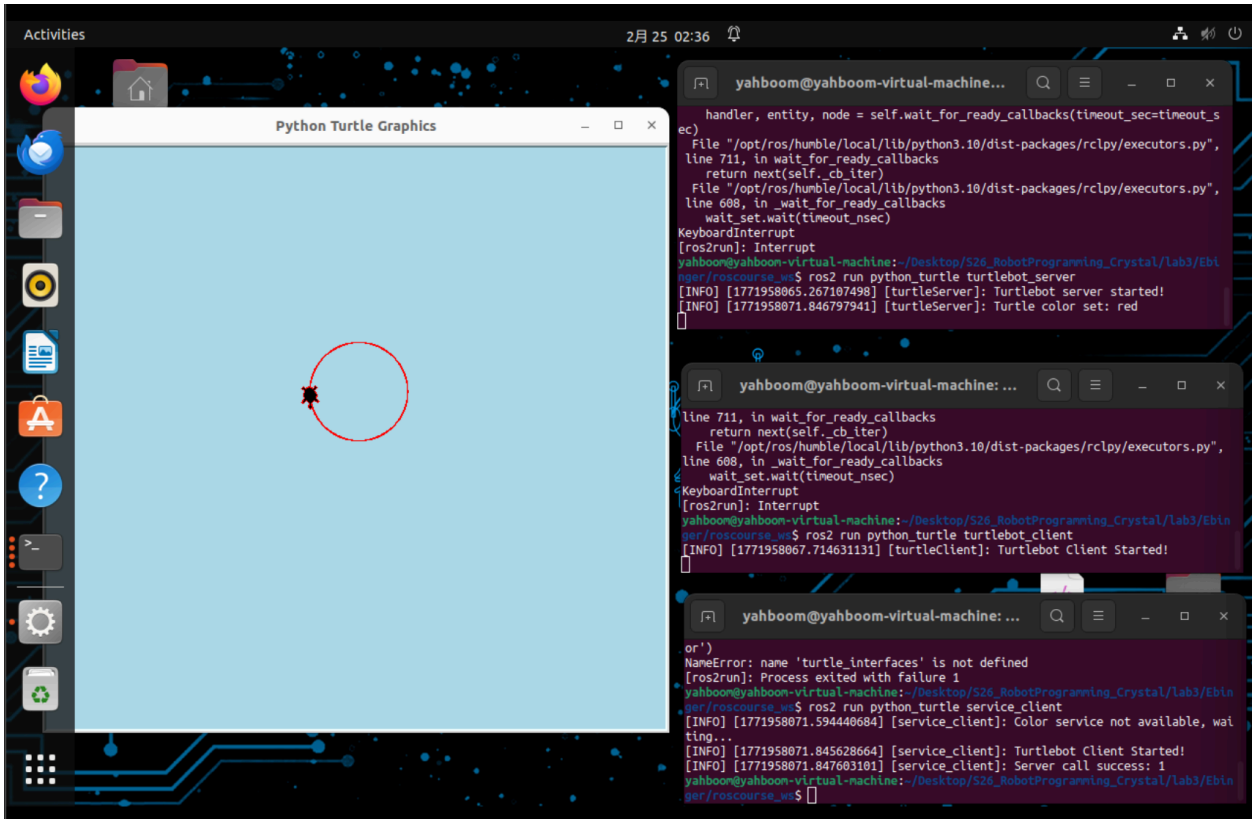
<u>Turtlebot_client:</u> The purpose of Turtlebot client is to create and maintain the simulation alongside the turtle objects inside of it. It has minimal computational qualities, instead calling upon the server to compute properties of the simulation instead. The class TurtleClient functions to display the simulation window and turtle to the user, update the simulation with information received from the server, and generate the publisher and subscriber nodes that transfer information between the server and client. Outside of that, the file also writes specific qualities to Turtlebot, specifically its color and position, if they are requested from the server. If not, it defines default values for these qualities.

Step 7



Question 8 - Once the server and client were both running, the turtle moved around in a small circle from the starting point.

Step 16


Activities
2月 25 02:36
Python Turtle Graphics
yahboom@yahboom-virtual-machine...
handler, entity, node = self.wait_for_ready_callbacks(timeout_sec=timeout_sec)
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/executors.py", line 711, in wait_for_ready_callbacks
return next(self._cb_iter)
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/executors.py", line 608, in _wait_for_ready_callbacks
wait_set.wait(timeout_nsec)
KeyboardInterrupt
[ros2run]: Interrupt
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Crystal/lab3/Ebinger/roscourse_ws$ ros2 run python_turtle turtlebot_server
[INFO] [1771958065.267107498] [turtleServer]: Turtlebot server started!
[INFO] [1771958071.846797941] [turtleServer]: Turtle color set: red
yahboom@yahboom-virtual-machine: ...
line 711, in wait_for_ready_callbacks
return next(self._cb_iter)
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/executors.py", line 608, in _wait_for_ready_callbacks
wait_set.wait(timeout_nsec)
KeyboardInterrupt
[ros2run]: Interrupt
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Crystal/lab3/Ebinger/roscourse_ws$ ros2 run python_turtle turtlebot_client
[INFO] [1771958067.714631131] [turtleClient]: Turtlebot Client Started!
yahboom@yahboom-virtual-machine: ...
or')
NameError: name 'turtle_interfaces' is not defined
[ros2run]: Process exited with failure 1
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Crystal/lab3/Ebinger/roscourse_ws$ ros2 run python_turtle service_client
[INFO] [1771958071.594440684] [service_client]: Color service not available, waiting...
[INFO] [1771958071.845628664] [service_client]: Turtlebot Client Started!
[INFO] [1771958071.847603101] [service_client]: Server call success: 1
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Crystal/lab3/Ebinger/roscourse_ws$